

Saarangsh Gupta

 saarangshgupta.com |  sgupta1_be22@thapar.edu |  [saarangshgupta19](https://www.linkedin.com/in/saarangshgupta19) |  [mesaarangsh](https://github.com/mesaarangsh)

EDUCATION

Thapar Institute of Engineering and Technology
Bachelor of Engineering in Computer Sciences (CGPA: 8.25)

Patiala, Punjab
Apr. 2022 – Apr. 2026

BCM Arya International
12th Grade - A levels (CGPA: 9.5)

Ludhiana, Punjab
Mar. 2020 – Mar. 2022

EXPERIENCE

Full Stack Intern

Apr 2024 – Feb 2025

ACT Center, Thapar School of Liberal Art and Sciences

Patiala, Punjab

- Developed a **NextJS** application to parse **BibTeX** files, dynamically displaying publications on **TIET's official website**.
- Implemented a **RESTful API** using **NodeJS** to enable efficient data retrieval for a Learning Management System.
- Built a **full-stack** web application using **React**, **NodeJS** and **PostgreSQL** to visualize GitHub commit activity, enabling real-time tracking of junior developers' contributions and improving team workflow efficiency by 30%.

Full Stack Intern

Aug 2024 – Dec 2024

PraxisForge Pvt. Ltd.

Remote | Delhi, India

- Authored 10,000+ lines of code to the company's repository, driving a 20% boost in system performance across domains.
- Pioneered an application to enable the review of news articles prior to publication decreasing false rate by 40%.
- Integrated **MongoDB** with a Flutter based application, leveraging **Redis** for caching frequently accessed data.

Undergraduate Research Project

Sep 2022 – Sep 2023

Pollution Control Board (PPCB), Ministry of Environment

Patiala | Chandigarh, India

- Ideated and scoped an industry institute interface to connect ITIs with industry, strengthening state-wide collaboration.
- Managed daily logs and automated communications using existing systems to enhance operational efficiency by over 40%.
- Analyzed operational workflows between the Pollution Control Board and Thapar Institute, pinpointing bottlenecks in data sharing and reducing reporting lag time by 15 days.

PROJECTS

BoxBox - Formula 1 Strategy Maker | *Python, NumPy, Pandas, React*

Feb 2025 – Mar 2025

- Conceived a tire degradation model using **regression** to predict tire usage across laps and different rubber materials.
- Programmed a race strategy simulator leveraging data from **FastF1 API** for prediction of optimal pit stops in a race.
- Optimized strategies using data modeling, and over **10,000 Monte Carlo** simulations aimed at race-time minimization.

EasyFind - Smart Lost & Found System | *NodeJS, React, Appwrite, Python*

Aug 2024 – Oct 2024

- Revamped Thapar University's lost-and-found system using Python-powered AI image captioning for efficient tracking.
- Improved owner identification, reducing search and wait times impacting over **2,000 users** across the campus.
- Engineered pipelines using CLIP and **sentence-transformers** for embedding both images and structured metadata, integrated with **Pinecone vector database** for real-time semantic increasing search efficiency to 90%.

Splitzee - Split Funds Easily | *NodeJS, React, MongoDB*

Jun 2024 – Jul 2024

- Developed core functionality and launched Splitzee for community, leading to 700+ weekly users within the first month.
- Incorporated **TanStack Query** for client-side caching, reducing traffic on backed services and boosting performance by **40%**.
- Implemented an intuitive UI, which enabled users to expedite expense splitting which resulted in a 95% user satisfaction rate.

ACHIEVEMENTS

- Spearheaded five tech initiatives as the Technical Director of **Google Developers Club**, including machine learning and web development projects, engaging **200+ students** across Thapar Institute of Engineering and Technology (TIET).
- Mentored 20+ Creative Computing Society (CCS) juniors, fueling over 10 competition victories across the campus.
- Represented Ludhiana at district level badminton championship in 2020.

SKILLS

Languages: JavaScript, Python, GoLang, C/C++, SQL, R

Frameworks: NextJS, ReactJS, NodeJS, Django, Appwrite

Developer Tools: Docker, Kubernetes, Jenkins, Git, Postman

Relevant Coursework: Data Structures(DSA), DBMS, CN, OOPS, OS, Data Science